THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 05/10/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

High online return rates exceeding 25% caused by consumers misjudging true furniture scale, dimensions, and fabric color within their physical home environments prior to purchasing.

2. AI RATIONALE & ANALYSIS

"Augmented reality directly solves spatial and chromatic uncertainty by overlaying true-to-scale, physically based rendered 3D digital furniture into physical customer interiors before checkout. THEXRA View provides instantaneous cross-platform spatial viewing without forcing users to download heavy native applications, ensuring rapid adoption within the 1-3 month timeline."

3. PROPOSED SOLUTION CONCEPT

A zero-friction, web and mobile-compatible augmented reality furniture visualization tool powered by THEXRA View, enabling customers to project photorealistic, true-to-scale 3D furniture models into their living spaces with dynamic color and fabric switching.

IMPLEMENTATION APPROACH:

1. Phase 1: Catalog audit, 3D asset pipeline definition, and WebXR player design.
2. Phase 2: Core e-commerce SDK embedding, material variation configuration, and automated scale validation.
3. Phase 3: Cross-device testing, analytics integration, production deployment, and telemetry tracking.

DETAILED IMPLEMENTATION PLAN:

1. **Discovery** (1 week):Technical architecture specification • 3D model optimization and LOD standards document • E-commerce platform integration plan
2. **Design** (1.5 weeks):AR UI/UX interaction wireframes • Product page 'View in AR' button integration specs • Color and fabric variant switcher design system
3. **Prototype** (1.5 weeks):Interactive prototype with top 5 high-return furniture SKUs • Spatial scale accuracy and plane detection validation report
4. **Development** (3 weeks):THEXRA View runtime integration into production staging • Dynamic 3D model asset ingestion pipeline • E-commerce cart event hook and analytics tracking script
5. **Testing** (1.5 weeks):Cross-platform iOS and Android QA matrix • Dimensional accuracy verification report • Web load time and asset compression benchmark
6. **Deployment** (1.5 weeks):Production deployment across primary product catalog • Live telemetry tracking verification • System administrator and merchandising handover package
7. **Support** (Ongoing):Monthly asset rendering performance audits • Return rate correlation and telemetry reports • Ongoing WebXR browser compatibility maintenance

SCOPE OF WORK (SOW)

1. PROJECT SCOPE

Delivery and deployment of THEXRA View AR furniture visualization embedded into the retailer's mobile web e-commerce journey to eliminate fit and color misjudgments.

2. CORE FEATURES

- Instant App-less WebXR and native QuickLook AR viewing
- 1:1 true-scale physical dimension rendering
- Real-time material, texture, and color variant switching

3. DELIVERABLES

- THEXRA View integration JavaScript SDK
- Optimized glTF and USDZ 3D model processing pipeline
- E-commerce analytics integration script
- Merchandising operations manual

- Direct room-to-cart checkout conversion trigger
- Real-time ambient light matching for realistic shadows and tone

4. HARDWARE REQUIREMENTS

- Standard client iOS devices (iPhone 8 and newer)
- Standard client Android devices (ARCore compatible)

5. SOFTWARE MODULES

- THEXRA View Spatial Runtime
- THEXRA Asset Optimization Engine
- WebXR Device API Integration Suite

6. CONTENT ASSETS

- Initial conversion and optimization of initial priority catalog 3D models
- PBR material library for standardized wood, leather, and fabric finishes

7. TRAINING PLAN

Conduct two remote training sessions for digital merchandising and catalog operations teams covering 3D asset upload and CMS management.

8. DEPLOYMENT STRATEGY

Containerized CDN script deployment embedded within customer-facing staging and production e-commerce environments.

9. ONGOING SUPPORT

Standard 99.9% uptime SLA for spatial asset hosting, ongoing browser engine update patches, and monthly analytics review.

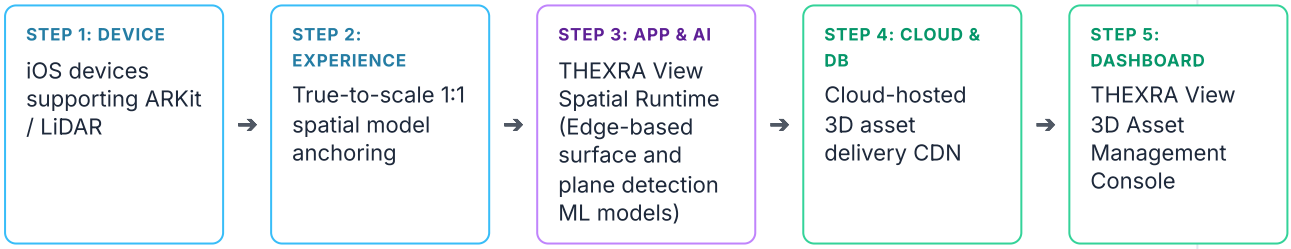
10. EXCLUSIONS (OUT OF SCOPE)

- Manual physical photogrammetry capture of furniture items on-site
- Redesign of the client's base e-commerce checkout architecture
- Custom dedicated mobile native application builds (iOS/Android native apps)

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

AR — THEXRA View

AUTOMATED SYSTEM FLOW DIAGRAM:



1. Overview: THEXRA View integrates lightweight WebXR and AR Quick Look/Scene Viewer spatial runtimes directly into the retailer's product display pages. Real-time floor plane detection scales and anchors photorealistic 3D models into the user's room environment. Integrated telemetry feeds spatial interaction duration and color selections back into enterprise merchandising analytics.

2. User Journey: Customer browses furniture item on mobile browser → Taps 'View in Your Room' → Camera scans floor surface → Photorealistic 3D model anchors in true scale → Customer adjusts colorways and inspects fit → Direct 'Add to Cart' action with pre-validated dimensions.

3. Hardware: iOS devices supporting ARKit / LiDAR, Android devices supporting ARCore, Standard client e-commerce host servers

4. Software: THEXRA View Spatial Runtime, WebXR Device API, glTF/USDZ 3D Asset Pipeline, E-commerce JavaScript SDK

5. AI Components: Edge-based surface and plane detection ML models, Ambient lighting estimation and automatic reflection generation algorithms

6. XR Components: True-to-scale 1:1 spatial model anchoring, Real-time PBR material and texture swapper, Interactive 6DOF model rotation and translation controls

7. Backend: Cloud-hosted 3D asset delivery CDN, RESTful Product Catalog & Variant Sync API, Session telemetry and conversion event ingestion service

8. Dashboard: THEXRA View 3D Asset Management Console, Spatial Engagement & Return Correlation Analytics Dashboard

> **9. Data Flow:** User Mobile Browser → CDN 3D Asset Retrieval → Local ARKit/ARCore Surface Anchor → In-Session Configurator → Telemetry Ingestion → Analytics Dashboard
>
> **Alternative Option:** Mobile/Web — THEXRA Make

5. EXPECTED BUSINESS OUTCOMES

Drastic reduction in returns from misjudged scale or color, higher online conversion rates, and increased customer purchasing confidence.

6. IMPLEMENTATION NEXT STEPS

Initiate a rapid 3D asset audit and technical integration workshop to test WebXR pipeline compatibility with existing e-commerce product pages.